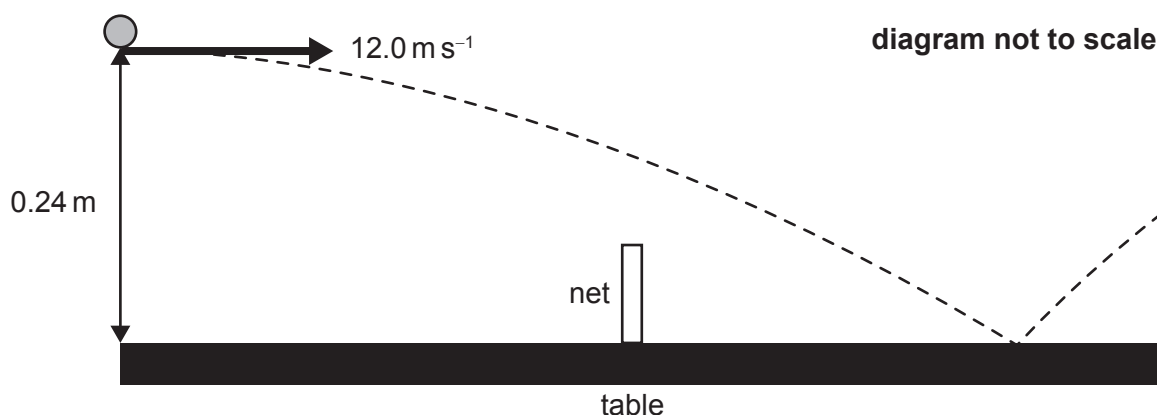


PAPER 2 / QUESTION 9

3 May 2021 / TZ1 / HL Paper 2 / original question 1 / 2221-6508

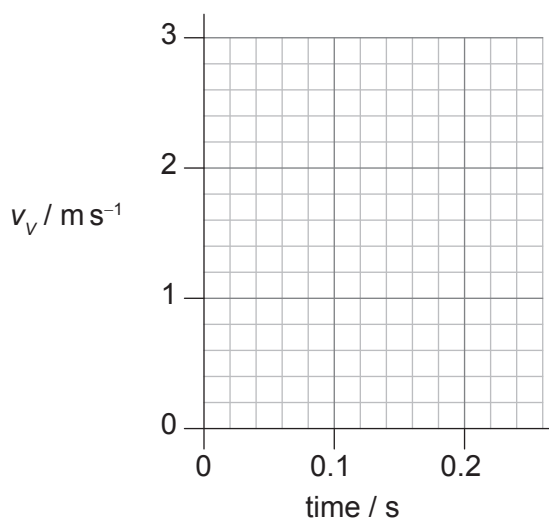
9. Two players are playing table tennis. Player A hits the ball at a height of 0.24 m above the edge of the table, measured from the top of the table to the bottom of the ball. The initial speed of the ball is 12.0 m s^{-1} horizontally. Assume that air resistance is negligible.



- (a) Show that the time taken for the ball to reach the surface of the table is about 0.2 s. [1]

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- (b) Sketch, on the axes, a graph showing the variation with time of the vertical component of velocity v_v of the ball until it reaches the table surface. Take g to be $+10 \text{ m s}^{-2}$. [2]



(This question continues on the following page)

(Question 9 continued)

- (c) The net is stretched across the middle of the table. The table has a length of 2.74 m and the net has a height of 15.0 cm.

Show that the ball will go over the net.

[3]

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- (d) The ball bounces and then reaches a peak height of 0.18 m above the table with a horizontal speed of 10.5 m s^{-1} . The mass of the ball is 2.7 g.

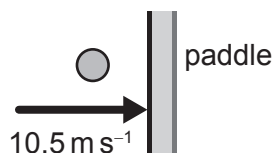
- (i) Determine the kinetic energy of the ball immediately after the bounce.

[2]

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- (ii) Player B intercepts the ball when it is at its peak height. Player B holds a paddle (racket) stationary and vertical. The ball is in contact with the paddle for 0.010 s. Assume the collision is elastic.



Calculate the average force exerted by the ball on the paddle. State your answer to an appropriate number of significant figures.

[3]

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